

Local convexity of nonlinear self-maps characterizes quadratic convexity

A candidate full solution to Problem 6.1 of arXiv:1205.3412

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Status: candidate full solution, likely valid. The property in Problem 6.1 forces the given norm to have modulus of convexity of power type 2. Consequently it also forces superreflexivity. Together with the forward theorem in the source paper, this gives an equivalence.

1 The question and the answer

For a Banach space X , Banach, Banach, Plichko and Prykarpatsky define

$$\text{conv}_2(X) = \inf_{\substack{x, y \in B_X \\ x \neq y}} \frac{1 - \|(x + y)/2\|}{\|x - y\|^2}.$$

This number is positive exactly when the given norm has modulus of convexity of power type 2. Their main theorem says that if $\text{conv}_2(X) > 0$, every Lipschitz-open, second-order Lipschitz map from an open subset of X to any Banach space is uniformly locally convex [1].

Problem 6.1 of the source asks for the converse when the codomain is X : if every Lipschitz-open, second-order Lipschitz map $f : U \rightarrow X$ is locally convex, must X have modulus of convexity of power type 2? Must it be (super)reflexive? We answer both questions affirmatively.

All arguments below are over the real scalar field. For a complex Banach space we apply them to its underlying real Banach space. This loses nothing: the hypotheses in the problem are metric, and convexity is real convexity.

Theorem 1. *For a Banach space X , the following are equivalent.*

1. $\text{conv}_2(X) > 0$.
2. *Every Lipschitz-open, second-order Lipschitz map $f : U \rightarrow X$, with $U \subset X$ open, is locally convex.*

If these conditions hold, then X is superreflexive (and hence reflexive).

The implication (1) $\Rightarrow$ (2) is the main theorem of the source paper. We prove the converse by contrapositive. The counterexample will be a quadratic perturbation of the identity on the unit ball.

2 Proof intuition

When $\text{conv}_2(X) = 0$, the unit ball has chords whose midpoint lies only $o((\text{chord length})^2)$ below a supporting hyperplane. A quadratic term of the natural chord-squared size can therefore push the preimage of an image midpoint outside a ball even when both endpoint preimages remain inside.

We realize these pushes simultaneously with a norm-convergent bilinear series. The possible rotation of the supporting functionals is the one nontrivial selection issue. Either some fixed output direction is seen by arbitrarily flat supports, or sufficiently flat supports become uniformly small on every prescribed finite set of directions. The first alternative gives a rank-one range; the second gives an almost diagonal series whose distinguished term dominates its past and tail. Finally, making the quadratic perturbation small compared with the identity guarantees Lipschitz openness and controls its inverse displacement.

3 Small quadratic perturbations are admissible

Lemma 2. *Let $B : X \times X \rightarrow X$ be a continuous symmetric real-bilinear map with $\|B\| \leq \kappa$, where $0 < 2\kappa < 1$. On $U = B_1(0)$ put*

$$f(x) = x + B(x, x).$$

Then f is injective, Lipschitz-open, and second-order Lipschitz. More precisely, with $L = 2\kappa$,

$$\|f(x) - f(y)\| \geq (1 - L) \|x - y\| \quad (x, y \in U), \quad (1)$$

$$\text{Lip}_o(f) \geq 1 - L, \quad (2)$$

$$\text{Lip}_2(f) \leq 2\kappa. \quad (3)$$

Proof. Write $Q(x) = B(x, x)$. For $x, y \in U$,

$$Q(x) - Q(y) = B(x - y, x) + B(y, x - y),$$

so $\|Q(x) - Q(y)\| \leq L \|x - y\|$. This proves (1), hence injectivity.

Suppose $B_r(x) \subset U$ and $\|w - f(x)\| < (1 - L)r$. Choose $s < r$ so that $\|w - f(x)\| < (1 - L)s$. On the complete closed ball $B_s(x)$, the map

$$T(z) = w - Q(z)$$

is an L -contraction and maps the ball strictly into itself, since

$$\|T(z) - x\| \leq \|w - f(x)\| + \|Q(x) - Q(z)\| < (1 - L)s + Ls = s.$$

Its fixed point belongs to $B_s(x)$ and satisfies $f(z) = w$. This proves (2). Finally,

$$f(x + h) - 2f(x) + f(x - h) = 2B(h, h),$$

which gives (3) with the definition used in the source. □

4 A bilinear form detecting arbitrarily flat midpoints

Assume henceforth that $\text{conv}_2(X) = 0$. For $x, y \in B_X$, set

$$z = \frac{x + y}{2}, \quad h = \frac{x - y}{2}, \quad d = 1 - \|z\|.$$

Call (x, y, σ) a *flat triple* if $x \neq y$, $z \neq 0$, $\sigma \in S_{X^*}$, and $\sigma(z) = \|z\|$. Its flatness ratio is $d/\|h\|^2$. Because

$$\inf \frac{d}{\|h\|^2} = 4 \operatorname{conv}_2(X) = 0,$$

flat triples with arbitrarily small ratio exist.

Set

$$\alpha_n = 16^{-n}, \quad \kappa = \sum_{n=1}^{\infty} \alpha_n = \frac{1}{15}, \quad C = (1 - 2\kappa)^{-1}.$$

We shall construct flat triples (x_n, y_n, σ_n) , vectors $v_n \in S_X$, functionals $\varphi_n \in S_{X^*}$, positive numbers β_n , and a symmetric bilinear map B such that

$$\|B\| \leq \kappa, \quad \sigma_n(B(h_n, h_n)) \geq \beta_n \|h_n\|^2, \quad (4)$$

where $h_n = (x_n - y_n)/2$. The only issue is that the norming functionals σ_n may rotate. The following exhaustive dichotomy handles it.

Case I: one direction is seen infinitely often

Suppose there are $v \in S_X$, $c > 0$, and flat triples of arbitrarily small ratio for which $|\sigma(v)| \geq c$. By passing to one sign along a sequence whose ratios tend to zero, and replacing v by $-v$ if needed, we may require $\sigma_n(v) \geq c$. Put

$$\beta_n = c\alpha_n.$$

Choose $\varphi_n \in S_{X^*}$ with $\varphi_n(h_n) = \|h_n\|$, and define

$$B(u, w) = \sum_{m=1}^{\infty} \alpha_m \varphi_m(u) \varphi_m(w) v. \quad (5)$$

The series converges in operator norm and $\|B\| \leq \kappa$. Moreover, all its scalar summands are nonnegative, so

$$\sigma_n(B(h_n, h_n)) = \sigma_n(v) \sum_m \alpha_m \varphi_m(h_n)^2 \geq c\alpha_n \|h_n\|^2.$$

Case II: every fixed direction eventually disappears

Suppose Case I fails. Then for every $v \in S_X$ and every $c > 0$ there is a threshold $\theta > 0$ such that

$$\frac{1 - \|z\|}{\|h\|^2} < \theta \implies |\sigma(v)| < c \quad (6)$$

for every flat triple (x, y, σ) . The same assertion holds simultaneously for any finite list of directions, by taking the minimum of the corresponding thresholds.

We choose the triples inductively. Once $v_1, \dots, v_{n-1}$ have been chosen, use (6) and then choose the next triple sufficiently flat that

$$\sum_{m < n} \alpha_m |\sigma_n(v_m)| \leq \frac{\alpha_n}{4}. \quad (7)$$

Set

$$v_n = \frac{z_n}{\|z_n\|}, \quad \beta_n = \frac{\alpha_n}{2},$$

so $\sigma_n(v_n) = 1$. Again take $\varphi_n \in S_{X^*}$ with $\varphi_n(h_n) = \|h_n\|$, and now define

$$B(u, w) = \sum_{m=1}^{\infty} \alpha_m \varphi_m(u) \varphi_m(w) v_m. \quad (8)$$

Then $\|B\| \leq \kappa$. Since

$$\sum_{m>n} \alpha_m = \frac{\alpha_n}{15},$$

the distinguished n th term, (7), and the tail estimate give

$$\begin{aligned} \sigma_n(B(h_n, h_n)) &\geq \alpha_n \|h_n\|^2 - \|h_n\|^2 \sum_{m<n} \alpha_m |\sigma_n(v_m)| - \|h_n\|^2 \sum_{m>n} \alpha_m \\ &\geq \left(1 - \frac{1}{4} - \frac{1}{15}\right) \alpha_n \|h_n\|^2 \geq \frac{\alpha_n}{2} \|h_n\|^2. \end{aligned}$$

Thus (4) holds in both cases.

5 The images of arbitrarily small balls are nonconvex

There is a harmless order-of-choice point in the preceding construction which we now make explicit. In either case, before choosing the n th flat triple choose $\varepsilon_n > 0$ so that $\varepsilon_n \downarrow 0$, $\varepsilon_n < 1/4$, and

$$2C\kappa^2\varepsilon_n + C^2\kappa^3\varepsilon_n^2 \leq \frac{\beta_n}{2}. \quad (9)$$

Then choose the triple sufficiently flat that, in addition to the relevant case condition,

$$d_n < \frac{1}{4}\varepsilon_n\beta_n \|h_n\|^2. \quad (10)$$

This is possible in Case I by its definition, and in Case II by (6). Thus the B just constructed satisfies all the preceding estimates for these choices.

Apply Lemma 2 to this B and let

$$f(x) = x + B(x, x), \quad x \in B_1(0).$$

The map is Lipschitz-open and second-order Lipschitz. Consider

$$a_n = \varepsilon_n x_n, \quad c_n = \varepsilon_n y_n.$$

The midpoint of their images is

$$\begin{aligned} w_n &= \frac{f(a_n) + f(c_n)}{2} \\ &= f(\varepsilon_n z_n) + \varepsilon_n^2 B(h_n, h_n), \end{aligned} \quad (11)$$

where bilinearity was used in the second line.

Indeed $B_{1/2}(\varepsilon_n z_n) \subset B_1(0)$, and

$$\|w_n - f(\varepsilon_n z_n)\| \leq \kappa \varepsilon_n^2 < \frac{1 - 2\kappa}{2}.$$

Thus Lipschitz openness gives an $r_n \in B_1(0)$ with $f(r_n) = w_n$; injectivity makes it unique. Write

$$r_n = \varepsilon_n z_n + u_n.$$

The lower Lipschitz estimate (1) and (11) imply

$$\|u_n\| \leq C\varepsilon_n^2 \|B(h_n, h_n)\| \leq C\kappa\varepsilon_n^2 \|h_n\|^2. \quad (12)$$

Expanding the equation $f(r_n) = w_n$ gives

$$u_n + 2\varepsilon_n B(z_n, u_n) + B(u_n, u_n) = \varepsilon_n^2 B(h_n, h_n).$$

Consequently, using $\|z_n\|, \|h_n\| \leq 1$ and (12),

$$\begin{aligned} \|u_n - \varepsilon_n^2 B(h_n, h_n)\| &\leq 2\kappa\varepsilon_n \|u_n\| + \kappa \|u_n\|^2 \\ &\leq \varepsilon_n^2 \|h_n\|^2 (2C\kappa^2\varepsilon_n + C^2\kappa^3\varepsilon_n^2) \\ &\leq \frac{1}{2}\varepsilon_n^2\beta_n \|h_n\|^2. \end{aligned}$$

Together with (4), this yields

$$\sigma_n(u_n) \geq \frac{1}{2}\varepsilon_n^2\beta_n \|h_n\|^2. \quad (13)$$

Now put

$$\rho_n = \varepsilon_n + \frac{1}{4}\varepsilon_n^2\beta_n \|h_n\|^2.$$

Here $\beta_n \leq \alpha_1 = 1/16$, so $\rho_n < 1$ and this ball lies in the domain of f . Both a_n and c_n lie in the open ball $B_{\rho_n}(0)$, while (10) and (13) give

$$\begin{aligned} \|r_n\| &\geq \sigma_n(r_n) \\ &= \varepsilon_n(1 - d_n) + \sigma_n(u_n) \\ &> \varepsilon_n - \frac{1}{4}\varepsilon_n^2\beta_n \|h_n\|^2 + \frac{1}{2}\varepsilon_n^2\beta_n \|h_n\|^2 = \rho_n. \end{aligned}$$

Thus $r_n \notin B_{\rho_n}(0)$. Since f is injective, w_n has no other preimage in that ball. But w_n is the midpoint of $f(a_n)$ and $f(c_n)$, so

$$f(B_{\rho_n}(0))$$

is not convex. As $\rho_n \rightarrow 0$, the map f is not locally convex at the origin. This proves the contrapositive of (2) $\Rightarrow$ (1).

Finally, $\text{conv}_2(X) > 0$ makes the given norm uniformly convex. Hence X is superreflexive, and in particular reflexive. This completes the proof of Theorem 1.

6 Scope and novelty check

The construction answers both clauses of Problem 6.1 and does not require differentiability beyond the source's second-order Lipschitz hypothesis. It also strengthens the qualitative conclusion: failure of quadratic convexity is witnessed by one explicit quadratic perturbation of the identity whose images of arbitrarily small balls centered at zero are nonconvex.

Cheap indexes for the present run and a bounded title/keyword search found the source paper and later papers discussing local convexity, but no prior proof of this converse. This is not an exhaustive novelty certification. The flat-triple dichotomy and the nonconvexity calculation should receive independent specialist review.

References

- [1] I. Banakh, T. Banakh, A. Plichko, and A. Prykarpatsky, *On local convexity of nonlinear mappings between Banach spaces*, arXiv:1205.3412 (2012), <https://arxiv.org/abs/1205.3412>.